Canada’s Media Monitor: In-Depth Analysis of Political Trends from Online Media Sources

Introduction

This project is a comprehensive system designed to collect and analyze a wide array of local public news and social media posts. Its primary aim is to convert large volumes of raw news data into usable and meaningful analytics. This project is driven by the necessity to interpret and quantify the changing stories and opinions found in our vast online media landscape, a task that is increasingly important for understanding political events and social trends that influence public conversations and policy-making.

A feature of this project is the use of Large Language Models (LLMs), specifically OpenAI's GPT. The integration of LLMs is a response to their growing prominence and enhanced capabilities in recent years. In this project, LLMs serve a crucial role beyond just assisting with tasks; they revolutionize the process of extracting complex data, a challenge that was difficult to address with traditional techniques. With LLMs, efficiently mining detailed insights from extensive text data is now more feasible. Alongside, "Political Insights" employs web scraping techniques, utilizes a Flask-Python framework for backend development, incorporates a user-friendly frontend interface, and applies various data analytics algorithms. Additionally, it manages data storage and retrieval through effective database management, all of which work together to transform diverse news data into insightful and actionable information.

Architecture Design

The architecture design focuses on effectively data collection, storage, processing, and visualization, ensuring a seamless flow of information from raw data to insightful analytics. The architecture consists of four key components, each with a specific role and a carefully chosen technology stack to optimize performance. Here is a high level illustration of the architecture:

1. **Data Crawling System**: This project utilizes Scrapy, a sophisticated tool for extracting news from various online sources. It functions like an advanced net, systematically capturing news articles, essential for building our data repository.

2. **Backend Development**: The backbone of our system employs Flask, a Python-based framework, adept at handling the intricacies of data management and processing. The data is organized and stored using SQLite3, a database known for its efficiency and reliability, making it ideal for our storage needs.

3. **Data Analysis Techniques**: To decipher trends and patterns in news topics, I implement statistical methods such as Exponential Smoothing. These techniques enable us to analyze how news subjects evolve over time and across different regions, providing us with deeper insights.

4. **Frontend Development**: The user-facing side of the system is developed using Vue.js, a framework that aids in creating an intuitive and engaging interface. To enhance the user experience, I integrate various JavaScript libraries, which assist in presenting the data in a visually appealing and digestible format.

System Implementations

Data Crawling System-Data Collection and Scraping

- **Sources:** The project sources news data from media stack, municipality news sources, social media posts such as X, facebook, or instagram. Mediastack aggregates multiple news media sources, providing a rich and diverse dataset.

- **Web Scraping:** News data is collected using Scrapy, a powerful Python-based web scraping tool. Scrapy is used for its efficiency in extracting large amounts of data and its capability to handle complex web structures.

- **Data Structure**: The scraped data includes headlines, abstracts, lead paragraphs, full content, publication times, links, sources, API sources, word counts, keywords, and geographical information. This comprehensive structure is essential for detailed analysis. A database model based on this data structure is designed to store news data. I use SQlite3 here for new data storage.

The program workflow is illustrated in the following chart:

Data Analysis - LLM and Statistical Methods

- **Large Language Models (LLM)**: OpenAI's GPT model is employed for extracting key information from news data. I utilize prompt engineering and design two prompt templates for the insights extraction purpose. This involves designing specific prompts for topic extraction and political bias analysis in news content, leveraging the advanced capabilities of GPT.

- **Geographic Data and Trend Index**: Based on news data, geographic meta info will be extracted as a label. Trend Index will be modeled by a dynamic time interval to represent the trend of the news topic.

- **Trend Prediction**: The project applies statistical methods like Moving Average, Exponential Smoothing Models to analyze news topic trends over time and across different U.S. regions. These techniques help identify patterns and predict future trends in the data.
- **Data Storage**: All the data is also stored in SQlite3.

Backend Server - Data Hosting

- **Backend Development with Flask**: Flask, a Python web framework, is used to build the backend. This framework manages the data flow between the frontend and the database, and it supports data communication between the user and the backend server.

Frontend Development and Visualization

- **Vue.js Framework**: The frontend is developed using Vue.js, a progressive JavaScript framework known for its flexibility and ease of integration. Vue.js is chosen for its ability to create dynamic and responsive user interfaces.
- **Data Visualization**: Various JavaScript libraries are employed for data visualization, enabling dynamic and interactive presentations of the analyzed data. This includes time-series graphs, bubble charts, and heatmaps to represent topic trends, media biases, and hotspots.

Applications

The potential applications of this project extend to a broad range of online information analysis, with significant implications for understanding public sentiment and societal trends. One of the core functionalities is sentiment analysis, a branch of natural language processing (NLP) that discerns whether the data reflects positive, negative, or neutral sentiments. This technique is particularly valuable in monitoring and identifying potential instances of hate speech, offering insights into various perspectives on contentious social issues. Given Canada's diverse mosaic population, this project faces the unique challenge of capturing and interpreting the myriad of nuanced viewpoints emanating from different social strata. To tackle this, the project is designed to segment and scrutinize distinct sources of online information, enabling a more precise understanding of who is discussing specific issues and from which geographic or social locations these discussions originate. By doing so, the project aims to provide a comprehensive and multifaceted picture of public opinion and discourse, reflecting the complexity and diversity of views in Canada's pluralistic society.

Demonstration:

I have created a basic version of my project based on news reports from the US, here is the website: https://polinsight.xyz/index.html